```
from google.colab import files
uploaded = files.upload()
```

Choose Files Screenshot… 133747.png

Screenshot 2026-08-07 133747.png(image/png) - 142757 bytes, last modified: 8/7/2026 - 100% done

```
Saving Screenshot 2026-08-07 133747.png to Screenshot 2026-08-07 133747.png
```

Gemini

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Get the name of the uploaded file
image_filename = list(uploaded.keys())[0]

# Read the image from the uploaded dictionary
image_data = uploaded[image_filename]
image = cv2.imdecode(np.frombuffer(image_data, np.uint8), cv2.IMREAD_COLOR)

# Convert image to float32 for processing to avoid clipping of negative values
image_float = np.float32(image)

# Define the sharpening kernel (Laplacian filter)
kernel = np.array([[0, 1, 0],
                   [1, -8, 1],
                   [0, 1, 0]])
sharpened = cv2.filter2D(image, -1, kernel)

# Apply the filter to get the Laplacian response.
# Use ddepth=cv2.CV_32F to get float output and prevent clipping.
laplacian = cv2.filter2D(image_float, cv2.CV_32F, kernel)

# Sharpen the image by subtracting the Laplacian from the original image.
# This enhances edges. The alpha factor can be adjusted for stronger/weaker sharpenin
alpha = 1.0 # Sharpening strength factor
```

```
sharpened_float = image_float - alpha * laplacian

# Clip the values to be within [0, 255] and convert back to uint8
sharpened = np.clip(sharpened_float, 0, 255).astype(np.uint8)

cv2_imshow(image)
cv2_imshow(sharpened)
```